

LARYNX

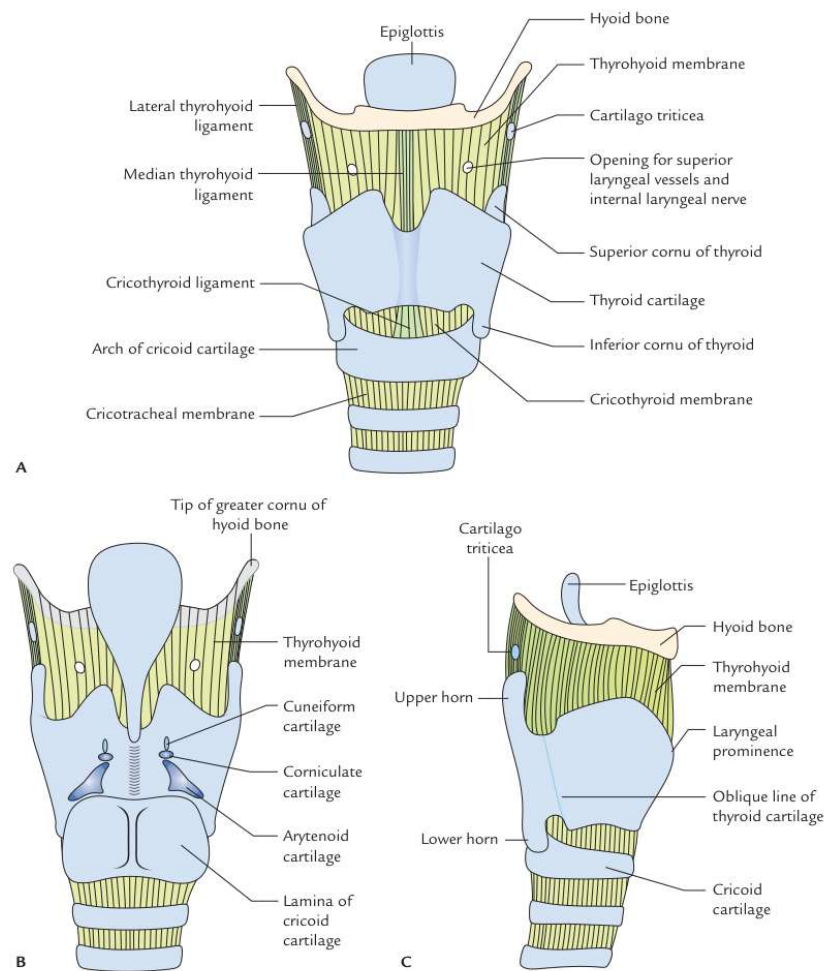
Regular Anatomy Notes | Gallbladder-style layout

Quick orientation

The larynx is the **upper expanded part of the lower respiratory tract** and is modified for voice production. It also works as a protective sphincter at the inlet of the lower respiratory tract.

It lies in the **anterior midline of the neck**, in front of the laryngopharynx, and extends from the root of the tongue to the trachea, opposite about **C3-C6 vertebrae**.

Primary role: **protection of lower respiratory tract**. Phonation is a later evolutionary function associated with speech.



Source diagram: skeleton of the larynx in anterior, posterior and lateral views.

How to read the skeleton diagram

The thyroid cartilage forms the anterior shield, the cricoid forms a complete ring below it, and the arytenoids sit on the cricoid lamina posteriorly.

The diagram also shows the thyrohyoid membrane, cricothyroid membrane and cricotracheal membrane, which are important for attachments and clinical procedures.

1. Functions, situation and size

Point	High-yield notes
Functions	Phonation , respiration, protection and deglutition. During swallowing the larynx moves upward and forward, helping to protect the airway.
Situation	Anterior midline of upper neck, in front of laryngopharynx. It extends from the root of tongue to trachea.
Vertebral level	Opposite C3, C4, C5 and C6 vertebrae; in children and females it lies a little higher.
Sex difference	The male larynx enlarges rapidly at puberty. The thyroid angle becomes prominent, forming the Adam's apple , and the voice becomes deeper.
Key clinical point	If the protective sphincter action fails, food or fluid may be aspirated into the trachea, causing choking or aspiration.

Easy memory line

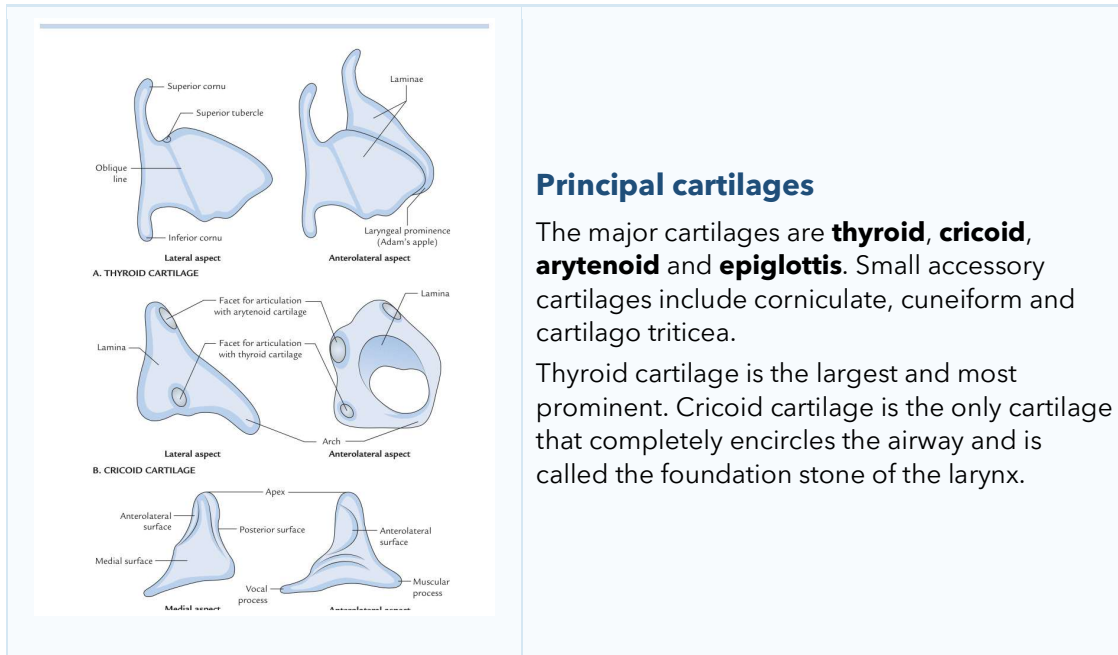
Larynx = Voice + Valve. It produces voice, but its more basic job is to guard the airway like a valve at the lower respiratory tract inlet.

Small clinical orientation

On the front of the neck, the laryngeal prominence of the thyroid cartilage is felt as Adam's apple, especially in adult males. Inferior to it lies the cricoid cartilage, and below cricoid the tracheal rings begin.

In viva, always link the level **C6** with the cricoid cartilage, beginning of trachea and the lower end of the pharynx/upper oesophagus region.

2. Cartilages of the larynx



Principal cartilages

The major cartilages are **thyroid**, **cricoid**, **arytenoid** and **epiglottis**. Small accessory cartilages include corniculate, cuneiform and cartilago triticea.

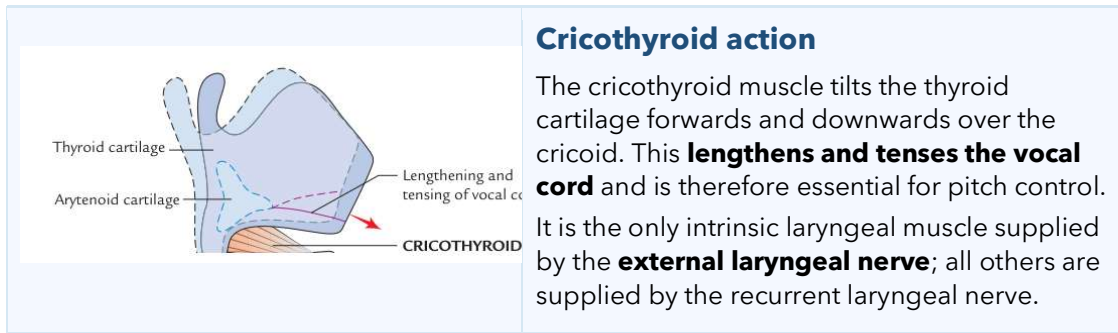
Thyroid cartilage is the largest and most prominent. Cricoid cartilage is the only cartilage that completely encircles the airway and is called the foundation stone of the larynx.

Cartilage	Important features
Thyroid cartilage	Largest cartilage. Made of two laminae meeting anteriorly at the thyroid angle. The posterior border is prolonged into superior and inferior cornua. Its laryngeal prominence is prominent in males.
Cricoid cartilage	Signet-ring shaped. Narrow anterior arch and broad posterior lamina. Situated at C6 and completely encircles the laryngeal lumen.
Epiglottis	Leaf-like elastic cartilage. Its broad upper end is free and forms the upper boundary of laryngeal inlet. Its stalk is attached to the thyroid angle by the thyroepiglottic ligament.
Arytenoid cartilages	Paired pyramidal cartilages on the cricoid lamina. Each has a vocal process anteriorly and a muscular process laterally/backwards. These are vital for vocal cord movement.
Corniculate and cuneiform	Small cartilages in the aryepiglottic fold region. They support the posterior part of the laryngeal inlet.

3. Joints of the larynx

Joint	Type and movements	Clinical / functional importance
Cricothyroid joint	Plane synovial joint between inferior cornu of thyroid cartilage and side of cricoid cartilage. Allows rotation and limited gliding.	Tilting at this joint changes length and tension of the vocal cords. Recurrent laryngeal nerve enters larynx close to this joint.

Joint	Type and movements	Clinical / functional importance
Cricoarytenoid joint	Tiny plane synovial joint between base of arytenoid and upper border of cricoid lamina. Allows rotation and gliding.	Most important joint for opening and closing the rima glottidis by abduction/adduction of vocal cords.
Arytenocorniculate joint	Synovial joint between arytenoid and corniculate cartilages.	No major functional importance; included for completeness.



4. Ligaments and membranes

How to classify them

Ligaments and membranes are described as **extrinsic** and **intrinsic**. Extrinsic structures connect the larynx to neighbouring structures. Intrinsic structures form the fibroelastic framework inside the laryngeal cavity.

Group	Membranes / ligaments	Exam-useful notes
Extrinsic membranes	Thyrohyoid membrane and cricotracheal membrane.	Thyrohyoid membrane is pierced by internal laryngeal nerve and superior laryngeal vessels. Cricotracheal membrane connects cricoid to first tracheal ring.
Extrinsic ligaments	Median and lateral thyrohyoid ligaments, cricothyroid ligament, cricotracheal ligament, thyroepiglottic and hyoepiglottic ligaments.	Cartilago triticea may occur in the lateral thyrohyoid ligament.
Intrinsic membrane	Cricovocal membrane / conus elasticus and quadrangular membrane.	Conus elasticus thickens superiorly to form the vocal ligament. Quadrangular membrane contributes to vestibular ligament.
Important free edges	Vestibular ligament and vocal ligament.	Vestibular ligament forms false vocal cord; vocal ligament forms the core of true vocal cord.

5. Muscles of the larynx

Intrinsic muscles

Intrinsic muscles move laryngeal cartilages and act mainly on the laryngeal inlet and vocal folds. Their actions open/close the inlet, abduct/adduct vocal cords and alter vocal cord tension.

A key exam point: **posterior cricoarytenoid** is the only abductor of vocal cords and is therefore called the safety muscle of the larynx.

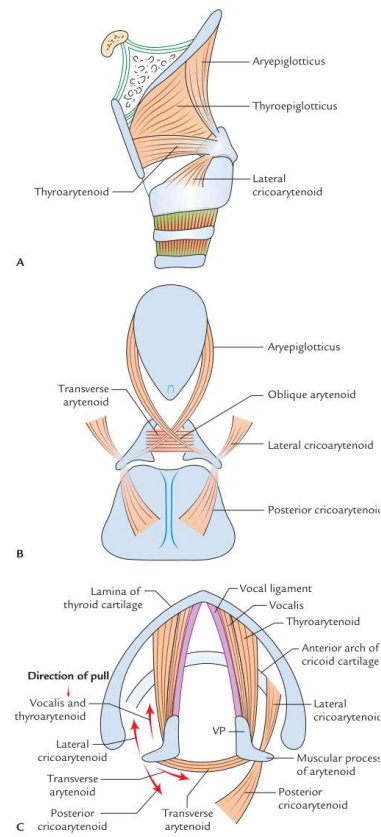


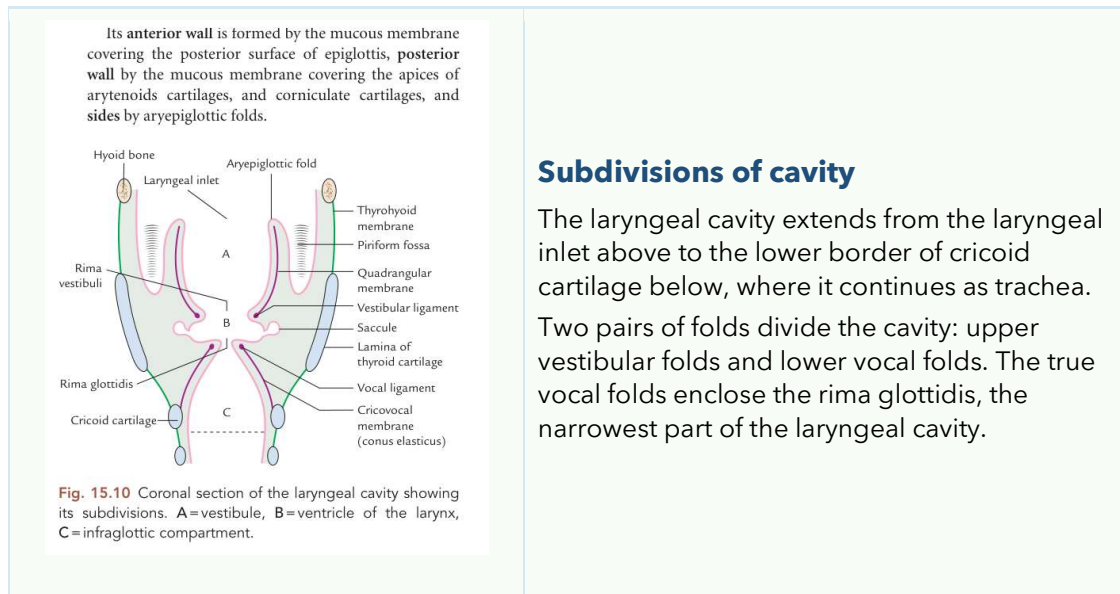
Fig. 15.6 Intrinsic muscles of the larynx: A, lateral view

Action group	Muscles	Main result
Open/close laryngeal inlet	Oblique arytenoid and aryepiglotticus close inlet; thyroepiglotticus opens inlet.	Regulates entry into laryngeal cavity during swallowing and breathing.
Abduct vocal cords	Posterior cricoarytenoid.	Opens rima glottidis; essential for breathing.
Adduct vocal cords	Lateral cricoarytenoid and transverse arytenoid.	Closes rima glottidis; needed for phonation and protective closure.
Increase tension	Cricothyroid and vocalis.	Produces tense cords; important in pitch adjustment.
Decrease tension	Thyroarytenoid.	Relaxes vocal cords and helps modulation of voice.

Nerve supply rule for muscles

All intrinsic muscles of larynx are supplied by recurrent laryngeal nerve except cricothyroid, which is supplied by the external laryngeal nerve.

6. Cavity of the larynx



Part of cavity	Extent	Main points
Vestibule (supraglottic compartment)	From laryngeal inlet to vestibular folds.	Separated from piriform recesses by aryepiglottic folds.
Ventricle / sinus of larynx	Between vestibular and vocal folds.	Its saccule contains mucous glands that lubricate vocal cords; called the oil can of larynx.
Infraglottic compartment	From vocal folds to lower border of cricoid cartilage.	Continues inferiorly with trachea.

Laryngeal inlet boundaries

Anterior: epiglottis. **Posterior:** interarytenoid fold of mucous membrane. **Lateral:** aryepiglottic folds on each side.

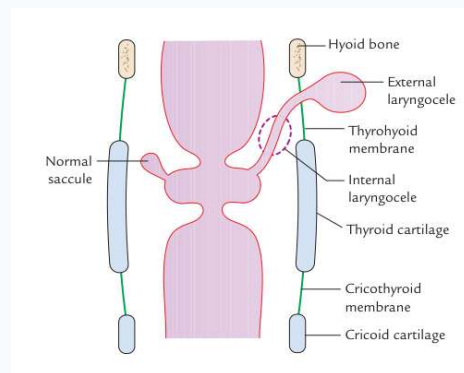
7. Mucous membrane and neurovascular supply

Structure	Details
Mucosa	Loosely attached except over posterior surface of epiglottis, true vocal cords, corniculate and cuneiform cartilages, where it is firmly adherent.
Epithelium	Mostly ciliated columnar epithelium. Vocal folds, upper parts of aryepiglottic folds and parts of epiglottis are lined by stratified squamous epithelium.
Mucous glands	Present throughout mucosa, especially posterior epiglottis, posterior aryepiglottic folds and saccules. No mucous glands in vocal folds.
Motor nerve supply	Internal laryngeal and external laryngeal nerves are

Structure	Details
	described in the source; functionally, cricothyroid is external laryngeal, other intrinsic muscles are recurrent laryngeal.
Sensory nerve supply	Above vocal folds: internal laryngeal nerve. Below vocal folds: recurrent laryngeal nerve.
Arterial supply	Above vocal fold: superior laryngeal artery from superior thyroid artery. Below vocal fold: inferior laryngeal artery from inferior thyroid artery.
Venous drainage	Superior laryngeal vein drains to superior thyroid vein; inferior laryngeal vein drains to inferior thyroid vein.
Lymph drainage	Above vocal cords: through thyrohyoid membrane to upper deep cervical nodes. Below vocal cords: through cricothyroid membrane to prelaryngeal and pretracheal nodes, then lower deep cervical nodes.

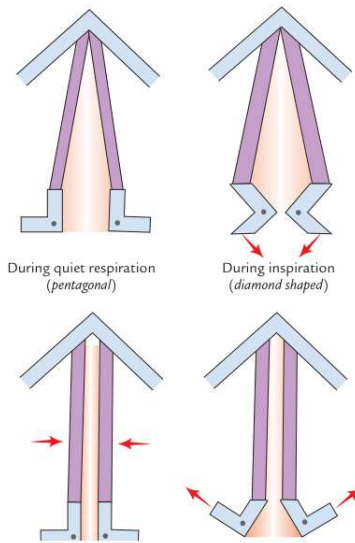
Clinical image: laryngocele

A laryngocele occurs when the laryngeal saccule dilates due to raised air pressure, as in trumpet players, glass blowers or weight lifters. It may be internal within the larynx or external when it herniates through the thyrohyoid membrane.



8. Rima glottidis and phonation

Rima glottidis is divided into the following two parts:
Intermembranous part in the anterior three-fifths between the vocal cords.



Shapes of rima glottidis

Rima glottidis is the narrowest anteroposterior cleft of the laryngeal cavity. It is bounded anteriorly by thyroid angle, posteriorly by interarytenoid fold and laterally by vocal folds and vocal processes.

Its anterior three-fifths is the intermembranous part and posterior one-fifth is the intercartilagenous part.

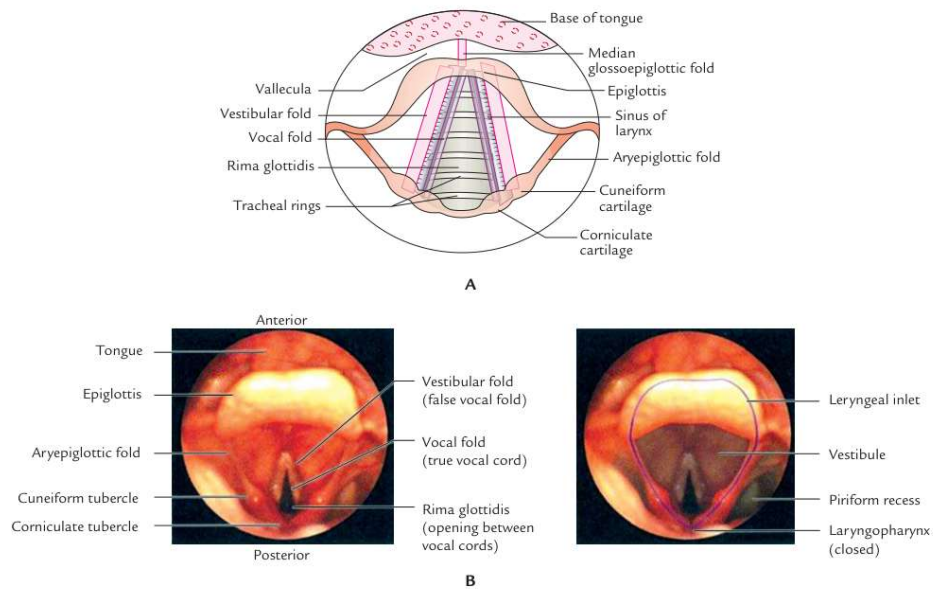
Condition	Shape of rima glottidis
Quiet breathing	Intermembranous part triangular; intercartilagenous part rectangular. Whole glottis appears pentagonal.
Full inspiration	Glottis widens and becomes diamond shaped due to abduction of vocal cords.
High-pitched voice	Rima glottidis becomes a narrow linear chink due to adduction of both parts.
Whispering	Intermembranous part highly adducted; intercartilagenous part remains open as a triangular gap, producing an inverted funnel shape.

Mechanism of phonation

Vocal cords are first kept adducted. Expired air from lungs builds infraglottic pressure and opens the cords in small puffs. As air passes through, the cords vibrate to produce sound. The sound is amplified by mouth, pharynx, oesophagus and nose, and shaped into speech by lips, tongue, palate, pharynx and teeth.

Loudness depends on amplitude of vibration; pitch depends on frequency of vibration. Male vocal cords are longer, so the voice is generally louder and lower pitched.

9. Laryngoscopy and applied anatomy



Source diagram: laryngoscopic view during moderate respiration.

What to identify in laryngoscopy

During laryngoscopy, the examiner can view the base of tongue, valleculae, epiglottis, aryepiglottic folds, piriform fossae, false vocal cords, true vocal cords and the sinus of larynx.

True vocal cords are pearly white and lie medial to the false vocal cords. In moderate respiration, rima glottidis remains widely open.

How to read the laryngoscopic view

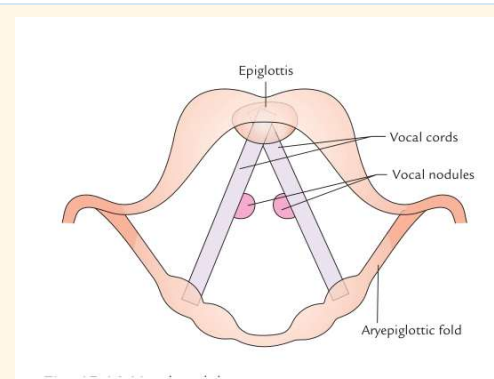
The epiglottis is anterior, while corniculate and cuneiform tubercles are posterior landmarks. The opening seen between the true vocal cords is the rima glottidis.

When answering viva, first identify epiglottis and vocal folds, then mention that true vocal cords are pale/white because their mucosa is tightly adherent and relatively avascular.

Vocal nodules

Vocal nodules are inflammatory nodules on the vocal cords, usually at the junction of anterior one-third and posterior two-third. They occur in people who overuse their voice, such as teachers and singers.

They are classically bilateral and symmetrical; early nodules are soft and reddish, later becoming whitish or greyish.



Applied anatomy	Important point
Laryngeal obstruction	Foreign body in the laryngeal inlet causes explosive coughing. If reflex is absent or slowed, choking and asphyxia can occur. Heimlich manoeuvre may expel the foreign body.
Internal laryngeal nerve injury	Loss of sensation above vocal folds and loss of protective cough reflex; foreign bodies may easily enter larynx.
Recurrent laryngeal nerve injury	Affects most intrinsic muscles. Bilateral injury may cause dangerous airway compromise because abductors are paralyzed.
External laryngeal nerve injury	Cricothyroid paralysis causes weak voice and inability to produce high-pitched notes.
Singer's nodules	At junction of anterior one-third and posterior two-third of vocal cord due to maximum friction.
Laryngeal carcinoma	Most common histological type is squamous cell carcinoma; glottic region is commonly involved.

10. Golden facts for revision

Question	Answer
Largest and most prominent cartilage	Thyroid cartilage
Cartilage that completely encircles laryngeal lumen	Cricoid cartilage
Foundation stone of larynx	Cricoid cartilage
Only abductor of vocal cords / safety muscle	Posterior cricoarytenoid
Chief tensor or tuning fork of larynx	Cricothyroid
Modulator of voice	Vocalis
Narrowest region of laryngeal cavity	Rima glottidis
Intrinsic muscle supplied by external laryngeal nerve	Cricothyroid
All other intrinsic muscles supplied by	Recurrent laryngeal nerve
Commonest congenital abnormality	Laryngomalacia
Reinke's space	Potential space under epithelium of vocal cord

Last-minute viva answer

The larynx is a cartilaginous, membranous and muscular organ in the anterior neck. It extends from root of tongue to trachea, opposite C3-C6. Its cartilages form the framework, intrinsic muscles move the vocal cords, and rima glottidis is the key functional opening for respiration and phonation.

Final checklist

Before closing the topic, revise these in order: **cartilages -> joints -> membranes -> intrinsic muscles -**

> cavity subdivisions -> rima glottidis -> nerve supply -> applied anatomy.

For diagrams, practise drawing the skeleton of larynx, coronal section of laryngeal cavity, and shapes of rima glottidis during breathing, inspiration, high-pitched voice and whispering.

One-line revision prompt	Answer to recall
Why is posterior cricoarytenoid called safety muscle?	It is the only abductor of vocal cords and keeps the airway open.
Why are true vocal cords white?	Their mucosa is firmly adherent to vocal ligament and has no intervening submucosa, giving an avascular pearly white appearance.
Why is cricoid clinically important?	It is a complete ring at C6 and forms a firm landmark at the lower end of the larynx.
Why can laryngocele become external?	The distended saccule may herniate through the thyrohyoid membrane.